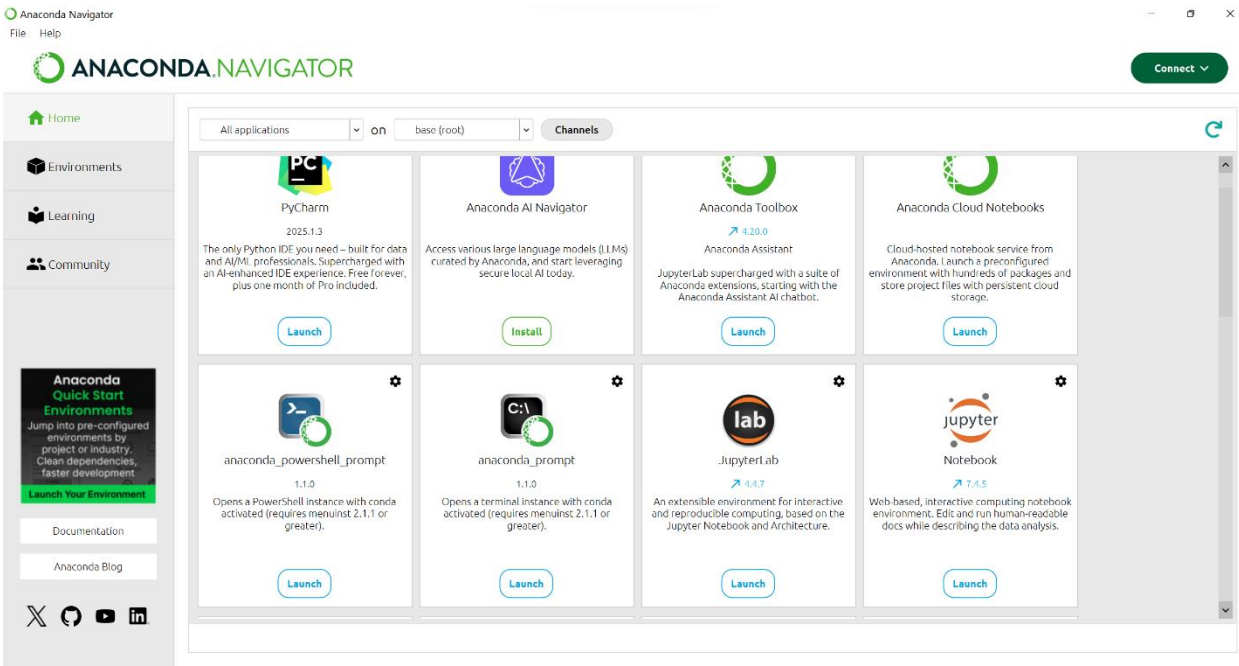
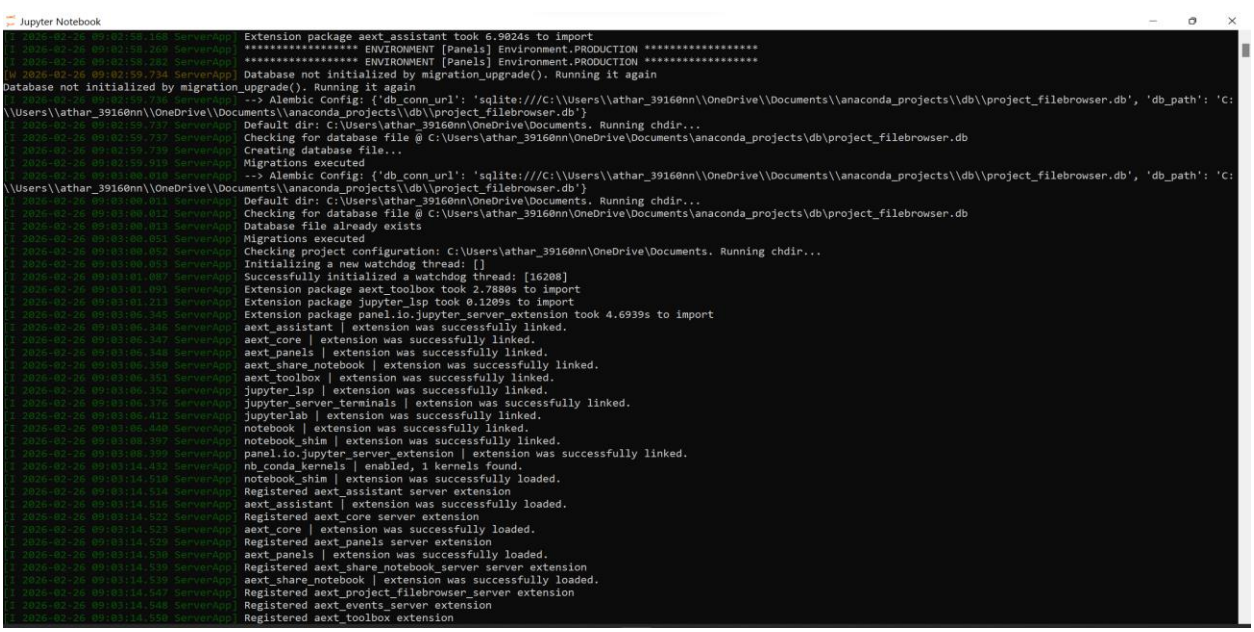
Name: Muhammad Athar Ur Rahman

Practical Tasks

1. Install Anaconda and verify Python installation.

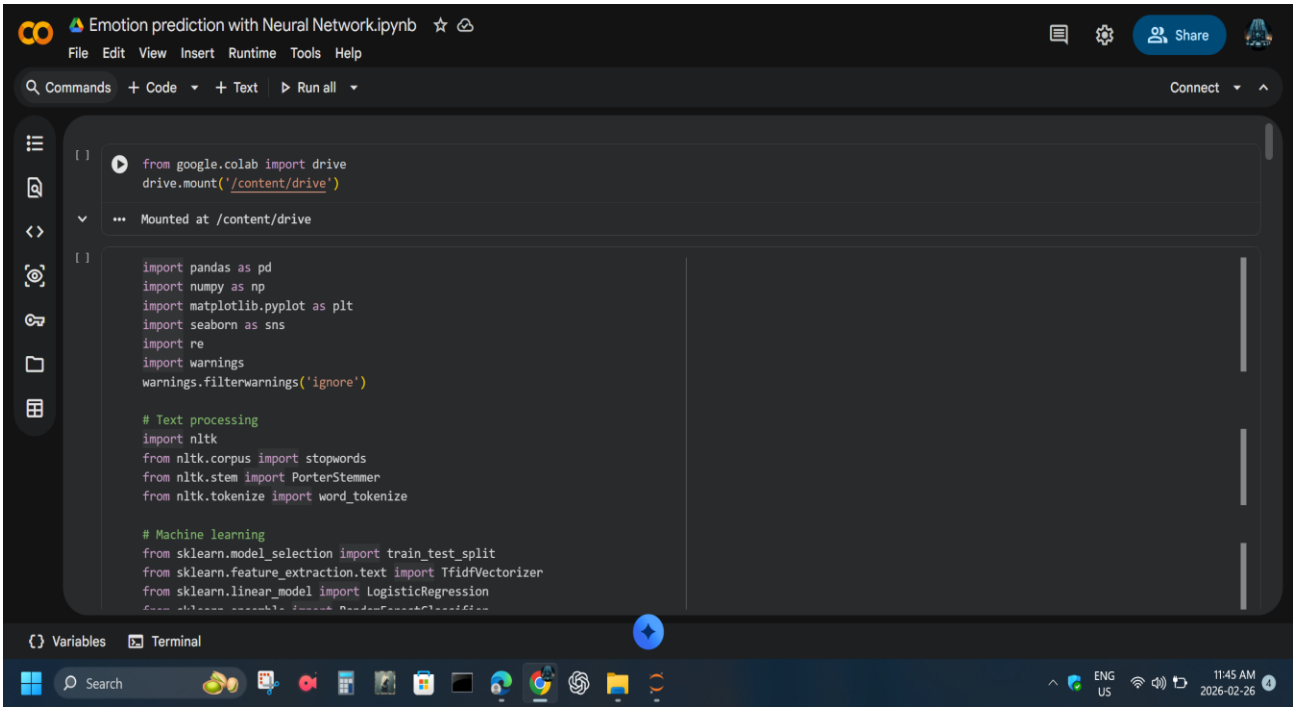


2. Set up Jupyter Notebook and Google Colab

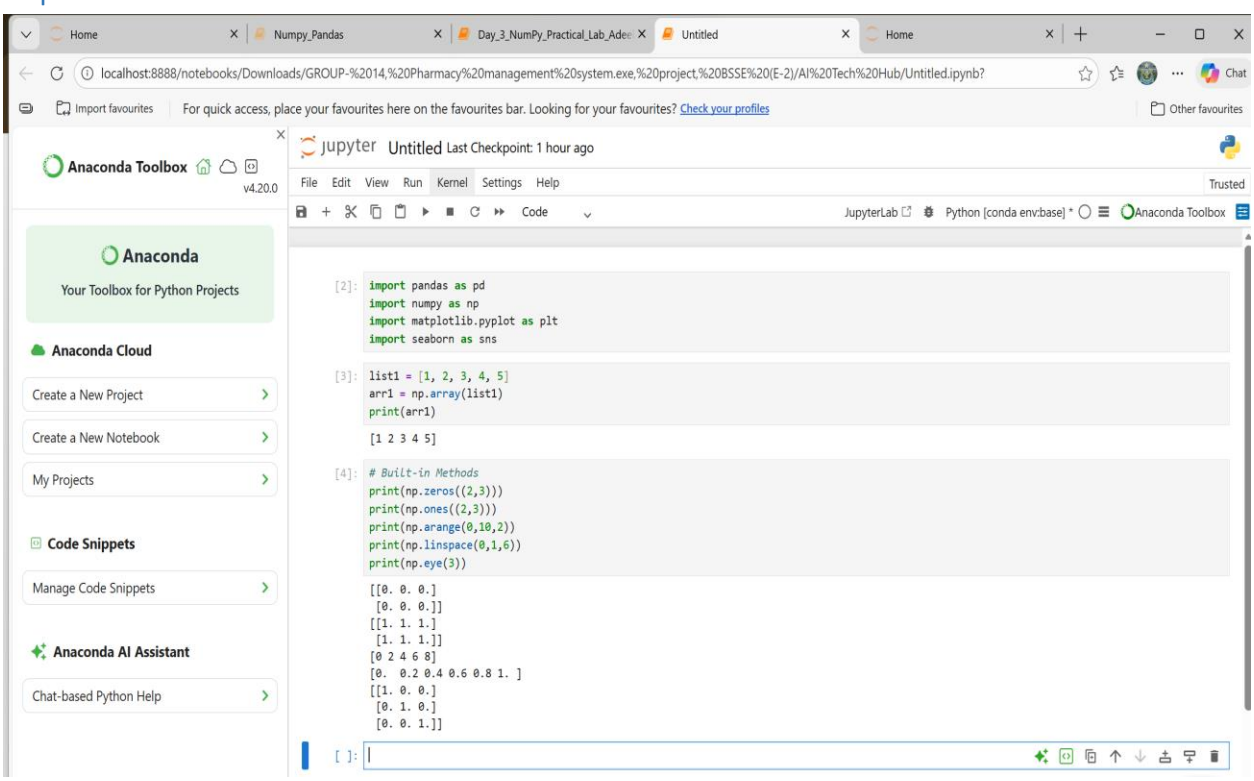


3. Install essential libraries:

Google Colab



Jupiter File



4. First Jupiter Notebook with basic Python operations

jupyter Untitled Last Checkpoint: 17 minutes ago

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Code JupyterLab Python [conda env:base] * Anaconda Toolbox

[1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

[2]:
```
df1 = pd.DataFrame({
    'ID': [1, 2, 3, 4],
    'Name': ['Ali', 'Sara', 'John', 'Adeel'],
    'Department': ['IT', 'HR', 'Finance', 'IT']
})

df2 = pd.DataFrame({
    'ID': [1, 2, 3, 5],
    'Salary': [50000, 60000, 55000, 70000]
})

df1, df2
```

[2]:
```
(   ID   Name Department
 0   1    Ali         IT
 1   2   Sara         HR
 2   3   John    Finance
 3   4  Adeel         IT,
    ID  Salary
 0   1   50000
 1   2   60000
 2   3   55000
 3   5   70000)
```

[3]:
```
pd.merge(df1, df2, on='ID', how='right')
```

[3]:

	ID	Name	Department	Salary
0	1	Ali	IT	50000
1	2	Sara	HR	60000
2	3	John	Finance	55000
3	5	NaN	NaN	70000

[4]:
```
pd.merge(df1, df2, on='ID', how='left')
```

[4]:

	ID	Name	Department	Salary
0	1	Ali	IT	50000.0
1	2	Sara	HR	60000.0
2	3	John	Finance	55000.0
3	4	Adeel	IT	NaN

[5]:
```
pd.merge(df1, df2, on='ID', how='right')
```

[5]:

	ID	Name	Department	Salary
0	1	Ali	IT	50000
1	2	Sara	HR	60000
2	3	John	Finance	55000
3	5	NaN	NaN	70000

[6]:
```
pd.merge(df1, df2, on='ID', how='outer')
```

[6]:

	ID	Name	Department	Salary
0	1	Ali	IT	50000.0
1	2	Sara	HR	60000.0
2	3	John	Finance	55000.0
3	4	Adeel	IT	NaN
4	5	NaN	NaN	70000.0

[10]:
```
data = {
    'Name': ['Ali', 'Sara', 'John'],
    'Age': [22, 25, 30],
    'Marks': [85, 90, 88]
}
df = pd.DataFrame(data)
df
```

[10]:

	Name	Age	Marks
0	Ali	22	85
1	Sara	25	90
2	John	30	88

[11]:
```
data = [
    ['Ali', 22, 85],
    ['Sara', 25, 90],
    ['John', 30, 88],
    ['Ahmed', 21, 78],
    ['Fatima', 23, 92],
    ['Usman', 27, 74],
    ['Ayesha', 24, 89],
    ['Bilal', 26, 81],
    ['Hina', 22, 95],
    ['Omar', 29, 67],
    ['Zain', 28, 84],
    ['Noor', 20, 91],
    ['Hamza', 23, 73],
    ['Iqra', 24, 86],
    ['Danish', 27, 79],
    ['Sana', 26, 94],
    ['Farhan', 25, 72],
    ['Mariam', 21, 88],
    ['Taha', 28, 83],
    ['Kiran', 22, 90]
]
df = pd.DataFrame(data, columns=['Name', 'Age', 'Marks'])
df
```

[11]:

	Name	Age	Marks
0	Ali	22	85
1	Sara	25	90
2	John	30	88
3	Ahmed	21	78
4	Fatima	23	92
5	Usman	27	74
6	Ayesha	24	89
7	Bilal	26	81
8	Hina	22	95
9	Omar	29	67
10	Zain	28	84
11	Noor	20	91
12	Hamza	23	73
13	Iqra	24	86
14	Danish	27	79
15	Sana	26	94
16	Farhan	25	72
17	Mariam	21	88
18	Taha	28	83
19	Kiran	22	90

[12]:
```
df['Name']
```

[12]:
```
0         Ali
1        Sara
2        John
3       Ahmed
4      Fatima
5       Usman
6      Ayesha
7       Bilal
8        Hina
9        Omar
10       Zain
11       Noor
12      Hamza
13       Iqra
14     Danish
15       Sana
16     Farhan
17     Mariam
18       Taha
19      Kiran
Name: Name, dtype: object
```

[13]:
```
df[['Name', 'Marks','Age']]
```

[13]:

	Name	Marks	Age
0	Ali	85	22
1	Sara	90	25
2	John	88	30
3	Ahmed	78	21
4	Fatima	92	23
5	Usman	74	27
6	Ayesha	89	24
7	Bilal	81	26
8	Hina	95	22
9	Omar	67	29
10	Zain	84	28
11	Noor	91	20
12	Hamza	73	23
13	Iqra	86	24
14	Danish	79	27
15	Sana	94	26
16	Farhan	72	25
17	Mariam	88	21
18	Taha	83	28
19	Kiran	90	22

[14]:
```
df.loc[7]
```

[14]:
```
Name      Bilal
Age          26
Marks        81
Name: 7, dtype: object
```

[15]:
```
df.iloc[0]
```

[15]:
```
Name      Ali
Age        22
Marks      85
Name: 0, dtype: object
```

Google Colab

```python
data=pd.read_csv("/content/drive/MyDrive/archive/tweet_emotions.csv")
```

```python
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 40000 entries, 0 to 39999
Data columns (total 3 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   tweet_id   40000 non-null  int64
 1   sentiment  40000 non-null  object
 2   content    40000 non-null  object
dtypes: int64(1), object(2)
memory usage: 937.6+ KB
```

```python
data.head()
```

	tweet_id	sentiment	content
0	1956967341	empty	@tiffanylue i know i was listenin to bad habi...
1	1956967666	sadness	Layin n bed with a headache ughhhh...waitin o...
2	1956967696	sadness	Funeral ceremony...gloomy friday...
3	1956967789	enthusiasm	wants to hang out with friends SOON!
4	1956968416	neutral	@dannycastillo We want to trade with someone w...

```python
data.tail()
```

	tweet_id	sentiment	content
39995	1753918954	neutral	@JohnLloydTaylor
39996	1753919001	love	Happy Mothers Day All my love

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